Fiscal Year:	FY 2006	**Task Last Updated:**	FY 08/16/2007
PI Name:	Kramerich, George L. Ph.D.		
Project Title:	Study of the Application of Contemporary Methods and Theories of Usage to Microgravity Activities		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid transport phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	216-687-2405
Organization Name:	Cleveland State University		
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City:	Cleveland	**State:**	OH
Zip Code:	44115	**Congressional District:**	11
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	11/03/2004	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Mullins, James M	**Contact Phone:**	216-433-8322
Contact Email:	James.M.Mullins@grc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NNC05GA04G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The main objectives of the Cleveland State University effort are to research and investigate the use of the latest technologies, methods, techniques, etc. which pertain to control and interface in industrial areas and study their application to research systems and facilities to improve efficiency, productivity, ergonomics, and other aspects of conceptualizing, executing, and using scientific experiments.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	Saving of money and time and improvement of usage in scientific endeavors.		

In the Power Systems Facility High Bay Area a major test facility of several hundred square feet called the Payload Rack Checkout Unit (PRCU) is in operation. There are six PRCUs at various NASA bases. The PRCU is an integration and test environment that provides high fidelity emulation of the data and resource interfaces between Payload Racks and the International Space Station. The PRCU uses the term “emulation” rather than “simulation” because actual hardware and Space Station Functional Equivalent Units are utilized to provide the physical resources (i.e., power, thermal cooling, gas, and vacuum) and the data resources to the payload rack under test. The PRCU, therefore allows the payload developer to complete development and verification prior to payload shipment to Kennedy Space Center for launch and integration on the International Space Station.
Cleveland State embarked on an in-depth continuing study of the PRCU, including the day-to-day operation, quality control, technical support systems, and training regimen for users. As part of the study, Cleveland State participated in the revamping of the extensive training regimen necessary for operators of the PRCU, resulting in significant cost and time savings. Previously it was necessary for all users to travel to Marshal Space Flight Center in Alabama for training on such systems as the Programmable Electrical Power System Emulator, Fire Detection System, Internal Thermal Control System, Vacuum System, Gas System, Internal Video Rack, Impedance Analysis Assembly, Command and Data Handling, and DAPC Delog System. The training is now done at Glenn Research Center by local qualified trainers on the PRCU that the users will be utilizing. When possible, the users are trained having their own equipment connected to the PRCU.

CSU was also closely involved in several other improvements in the design and operation of the PRCU. A study is being made of logistics and analysis methods involved in the analysis of Internal Temperature Control System fluid including tests for dissolved oxygen, phosphate, total organic carbon, chloride, metals, borate, silver, total digestion, and particle count. Comparisons are being made on efficiency and quality from in-house sources as compared to other sources.

The PRCU has been involved in the final design and testing of the Fluids and Combustion Facility (FCF), a section of the Space Station which includes the Combustion Integration Rack (CIR) and the Fluids Integration Rack (FIR). CSU is participating in studies in several areas of the handling and testing of these large Space Station facilities. CSU was also instrumental in improving methods of quality control and documentation of these projects. In the Power Systems Facility High Bay Area, where construction of International Space Station facilities takes place, Cleveland State is operating systems for the monitoring and ongoing study of activities and the environment, including the monitoring of air particle content, temperature, pressure, and humidity and the video monitoring and study of human activity in this environment.

All elements being built and tested for eventual use in the Space Station and Space Shuttle are tested for temperature and humidity performance in the Thermal Test Facility at Glenn Research Center. As a result of the study of the Thermal Test Facility, improvements suggested by Cleveland State for cutting time and costs of periodic calibration were implemented at a considerable savings of cost and down time in that facility. Cleveland State also studied issues of safety, ergonomics, and time constraints in the use of the Thermal Test Facility by such projects as the Fluids and Combustion Facility, Multi-user Droplet Combustion Apparatus, InSpace2, Light Microscopy Module, Image Processing and Storage Units, and others.

PROGRESS REPORT FOR FY 2006

Task Progress:

Cleveland State continued its in-depth study of the PRCU, including the day-to-day operation, quality control, technical support systems, and training regimen for users. As part of the study, Cleveland State continued its participation in the revamping of the extensive training regimen necessary for operators of the PRCU, resulting in significant cost and time savings. Previously it was necessary for all users to travel to Marshal Space Flight Center in Alabama for training on such systems as the Programmable Electrical Power System Emulator, Fire Detection System, Internal Thermal Control System, Vacuum System, Gas System, Internal Video Rack, Impedance Analysis Assembly, Command and Data Handling, and DAPC Delog System. The training is now done at Glenn Research Center by local qualified trainers using text and video materials created during this period on the PRCU that the users will be utilizing. When possible, the users are trained having their own equipment connected to the PRCU.

CSU also continued involvement in several other improvements in the design and operation of the PRCU. A study continues of logistics and analysis methods involved in the analysis of Internal Temperature Control System fluid including tests for dissolved oxygen, phosphate, total organic carbon, chloride, metals, borate, silver, total digestion, and particle count. It has been found that in-house sources of analysis and testing are more efficient and timely than the outsourcing of these activities. This process is now carried out at a savings of several thousands of dollars annually.

The PRCU’s primary purpose is the complete emulation of a rack connection to the Space Station. Until recently, this did not include emulation of the complete communications loop to ground control. CSU was instrumental in coordinating the upgrade of GRC’s communication in this regard. This included participation in design, installation, testing, and troubleshooting of equipment and systems. It was also necessary to integrate the new system into the GRC firewall and internet systems at GRC and Marshall Space Flight Center. In any test area, communications of problems and “lessons learned” greatly improves and streamlines present and future activities. In order to study the activities of PRCU personnel in this regard, CSU monitored and participated in the “Monthly User Forum” teleconferences of PRCU managers and users from GRC, Marshall Space Flight Center, Johnson Space Center, Kennedy Space Center, and other NASA installations.

Yearly calibration and maintenance of the PRCU subsystems is a vital part of the “emulation” role of this installation. This involves visiting personnel from several NASA installations, logistics involved in shipping equipment and supplies, coordination of activities, and large amounts of documentation. CSU monitored and participated in the entire process and made several proposals and took numerous actions to streamline the process and make it as user friendly as possible. The two-week process was finished three days early as a result.

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In the Power Systems Facility High Bay Area, where construction of International Space Station facilities takes place, Cleveland State is operating systems for the monitoring and ongoing study of activities and the environment, including the monitoring of air particle content, temperature, pressure, and humidity and the video monitoring and study of

	human activity in this environment. These activities continued this year with the continuing accumulation of data for eventual analysis. All elements being built and tested for eventual use in the Space Station and Space Shuttle are tested for temperature and humidity performance in the Thermal Test Facility at Glenn Research Center. As a result of the study of the Thermal Test Facility, improvements suggested by Cleveland State for cutting time and costs of periodic calibration were implemented at a considerable savings of cost and down time in that facility. Cleveland State also studied issues of safety, ergonomics, and time constraints in the use of the Thermal Test Facility by such projects as the Fluids and Combustion Facility, Multi-user Droplet Combustion Apparatus, InSpace2, Light Microscopy Module, Image Processing and Storage Units, and others.
Bibliography Type:	Description: (Last Updated:)